

Cobalt 60 is a man-made radioactive isotope used for industrial purposes to scan through thick materials like steel and concrete, as well as sterilizing medical equipment.

Pros

- Highly penetrating through the air
- Long 5.27 year half life to make it shelf-stable
- It has clearly defined peaks at 1.17 and 1.33 MeV, and can be directionalized through an opening to only make the radiation visible from a certain direction.

Cons

- Requires heavy shielding which makes it unsuitable for flight, its best use case would be for the “communications buoy” concept where a central node would collect information from a sector of buoys and transmit it up to the drone
- Gamma radiation is ionizing and very harmful

Wireless information transfer with fast neutrons

- Researchers used Californium-252 and successfully modulated the neutron flux field to encode binary data
- The sensors used liquid scintillation detectors to receive and then processed the signals
- Benefits included the ability to transmit through thick metal bulkheads, steel, and concrete. This could lead to use in high-security radioactive environments like submarines and nuclear facilities
- They are also not jammable by conventional EW
- Californium is extremely rare and expensive, as well as being extremely hazardous

Best use case would be L1 -> L2 communications where the buoy would be the one with the radioactive isotope and using a highly directionalized beam of the radiation to communicate to a drone.

MEet with cyber science librarian and email them beofre Friday 4th period